

HIGHLY SELECTIVE FISCHER-TROPSCH SYNTHESIS FOR C₁₀-C₂₀ DIESEL FUEL UNDER LOW PRESSURE

Zhen Dong,^{1,2} Haiping Zhang,¹ Tom Whidden,³ Ying Zheng^{1,4*} and Jianshe Zhao²

1. Department of Chemical Engineering, University of New Brunswick, Fredericton, E3B 5A3, NB, Canada

2. Department of Chemistry, Northwest University, Xian, 710069, Shaanxi, China

3. Atlantic Hydrogen Inc., 420 Wilsey Road, Fredericton, E3B 6E9, NB, Canada

4. Institute of Materials and Processes, School of Engineering, University of Edinburgh, Mayfield Road, Edinburgh, EH9 3JL, UK

This work deals with the improvement of diesel (C₁₀-C₂₀) production of Fischer-Tropsch (FT) synthesis at low pressure over Co-Re/Al₂O₃ since high catalytic activity and great potential in producing C₁₀-C₂₀ hydrocarbons were observed under conventional conditions. To improve the catalytic performance, CO conversion and hydrocarbon selectivity were investigated against pressure, syngas H₂/CO ratio, gas flow rate, and temperature. 93.7 % of CO conversion was achieved as well as 62.3 % diesel selectivity over Co-Re/Al₂O₃ catalyst even after working for 15 days, which was far higher than the maximum value of the Anderson-Schulz-Flory (ASF) distribution and the results in literature reports. It is suggested that Re plays an important role in enhancing dissociation of hydrogen as well as the formation of CH, CH₂, and CH₃ species to convert CO.

Keywords: Fischer-Tropsch synthesis, low pressure, diesel selectivity, catalytic conditions, Anderson-Schulz-Flory distribution

INTRODUCTION

Although petroleum-based fuels dominate the transportation market, the continued extraction and use of fossil fuels is highly problematic. Oil extraction faces problems with rapid depletion and more expensive extraction from existing reserves while continued use of fossil hydrocarbon-based fuels is the major contributor to anthropogenic climate change.^[1,2] These and other issues with the use of fossil hydrocarbon-based fuels make non-edible biomass attractive as a fuel resource. Biomass to liquid process is an alternative approach to transferring renewable energy from lignocelluloses to produce a clean liquid synthetic fuel, free from sulphur.^[3-5] Plasma, with great potential, is used as an external energy source to improve lignocelluloses mass yield of hydrogen and synthetic gas (or syngas) during biomass gasification.^[6,7] Plasma reactors primarily work at a gas pressure lower than 1000 kPa, which mismatches the high pressure requirement for the followed FT processes that typically operated at 2000–5000 kPa. Therefore, syngas generated from the plasma-enhanced gasification needs to be pressurized before being converted by FT synthesis, which is costly. To address this challenge, the present work investigates the FT synthesis at low gas pressures with an emphasis on production of diesel.

Although some catalysts especially iron and cobalt-based catalysts have been developed for FT synthesis, a few challenges still need to be overcome.^[8] With iron-based catalysts, the direction of producing lightweight olefinic hydrocarbons has been pursued, while research on cobalt-based catalysts has concentrated on the desire of diesel fuel and heavyweight products. Besides the above-mentioned pressure issue, selectivity control is considered as the other attractive challenge. Catalytic products over conventional FT catalysts typically follow the ASF distribution, generally multitudinous and unselective. The ASF distribution initially describes the relative ratios of products in the polymerization processes, based on chain growth probability (α). Apart from that, this distribution is also relevant to the FT process.

Lighter hydrocarbons are converted to heavier products through chain growth mode to produce a liquid synthetic fuel. According to ASF distribution equation, a maximum selectivity of 39.7 % to C₁₀-C₂₀ diesel fuels is obtained as α equals 0.87, which is not appreciated. Therefore, industrial FT synthesis with cobalt-based catalysts is always followed by the hydrocracking process to tune the selectivity to diesel.^[9] However, multiple steps will increase the economic cost and decrease the overall efficiency. From this point of view, the development of catalyst succeeded in the direct and selective production and further improving the FT technology becomes necessary and urgent. For example, bifunctional FT catalysts, which combine metal nanoparticles for CO hydrogenation and acid sites for hydrocracking, have been used to enhance the selectivity production of C₅-C₁₁.^[10] Recently, mesoporous zeolite-Y supported Co nanoparticles were reported to favour the formation of C₁₀-C₂₀ diesel fuel at 230 °C and 2000 kPa.^[11] Even though the results are good, some approaches and methodologies could be created to achieve higher conversion and product selectivity under even lower pressure.

In this article, Co-based catalysts supported on commercial mesoporous γ -Al₂O₃ were prepared and applied in FT synthesis aiming at increasing C₁₀-C₂₀ selectivity at low pressure. Compared to Co/Al₂O₃, Co-Re/Al₂O₃ showed some unique properties, e.g. high reducibility, good dispersion, and attractive catalytic performances. Additionally, the influences of operating conditions, e.g. pressure, H₂/CO ratio, gas hourly space velocity (GHSV, based on

* Author to whom correspondence may be addressed.

E-mail address: yzheng@unb.ca, y.zheng@ed.ac.uk

Can. J. Chem. Eng. 95:1537–1543, 2017

© 2017 Canadian Society for Chemical Engineering

DOI 10.1002/cjce.22812

Supplementary material is available in the online journal.

Published online 9 March 2017 in Wiley Online Library (wileyonlinelibrary.com).

catalyst volume), and temperature were investigated with the goal of the improvement of CO conversion and C₁₀-C₂₀ selectivity.

EXPERIMENTAL

Catalyst Preparation

Co-Re/Al₂O₃ with cobalt loading of 20 g/g and rhenium of 0.5 g/g supported on γ -Al₂O₃ (surface area: 148 m²/g, average pore size: 24.44 nm) was prepared through incipient co-impregnation of cobalt (II) nitrate hexahydrate (Aldrich) and ammonium perrhenate (Aldrich) onto the γ -Al₂O₃. The wet samples were allowed to dry at 120 °C overnight. The obtained samples were then pressed into tablets and further calcined for 6 h at 500 °C. The calcined tablets were crushed to small particles (0.18–0.25 mm in diameter) and further evaluated by application in a fixed-bed reactor.

Characterization

The samples were characterized by X-ray powder diffraction (XRD), N₂ adsorption-desorption isotherms, H₂ temperature-programmed reduction (TPR), transmission electron microscopy (TEM), and X-ray photoelectron spectroscopy (XPS). XRD patterns were obtained using a Bruker D8 Advance spectrometer using CuK α radiation (40 kV, 30 mA). TEM measurements were carried out with a JEOL JEM-200CX instrument at an accelerating voltage of 200 kV. H₂ TPR was recorded using a homemade reactor equipped with an MS detector. 100 mg of catalyst was pretreated at 200 °C in Ar flow (30 mL/min) for 1.5 h to remove moisture and then cooled down. The H₂ TPR curve was recorded until a stable baseline was reached using H₂/Ar (5 % H₂ + 95 % Ar, 20 mL/min). Cobalt species on the catalysts surface were determined by XPS (ESCALAB 250Xi XPS) equipped with a monochromatic Al K α X-ray source. Binding energies were calibrated using C1s peak (284.6 eV) as a reference.

Experimental Conditions

Catalysts were firstly evaluated in a 25 mL batch-type autoclave. 0.1 g catalyst precursor was reduced ex-situ for 6 h at 400 °C with H₂ (30 mL/min) and a heating rate of 1 °C/min, followed by in-situ reactivated at 400 °C, 2000 kPa H₂ atmosphere for 4 h and then stabilized at 220 °C. The reactor was fed with syngas (H₂/CO = 2.0) to 2000 kPa. After 12 h the gaseous products were characterized using a gas chromatograph, and liquid products were evaluated using GC/MS (Shimadzu GC/MS-QP 5000). Sampling of liquid products in batch reaction was carried out by dissolving and dispersing the products with catalyst in toluene. Studies on

catalytic activity under low pressure were performed on a fixed-bed reactor (Autoclave Engineering, BTR-Jr-PC; Figure 1). 2.0 g (2.0 mL) of unreduced catalyst (0.18–0.25 mm in diameter) was diluted using glass beads (0.15–0.18 mm in diameter) with a volume ratio of 1:1. The catalyst was reduced in-situ using the procedure described above and aged in flowing H₂/CO (2:1 L/L; 1000 h⁻¹) for 24 h at 300 kPa and 170 °C before data were recorded.

RESULTS AND DISCUSSION

Physicochemical Properties

Commercial alumina shows the diffraction pattern of γ -Al₂O₃ phase. Cobalt catalysts loaded on Al₂O₃ show similar crystalline patterns attributed to Co₃O₄ (Figure 2). The calculated crystal size by Scherrer's equation is about 17.2 nm for Co/Al₂O₃ and a decrease to 14.1 nm for Co-Re/Al₂O₃ is found. According to the literature, promotion by Re would favour the dispersion of metallic particles due to the increased amount of cobalt oxide crystal grain, thus generating a large number of small particles of cobalt species.^[12] Presumably, due to the low loading of Re, such a decrease is inconspicuous.

The evolution of morphology for the reduced catalysts is observed with TEM. The catalysts were constructed of nanoparticles dispersed on support (Figure 3). Particle growth during reduction occurs since some big particles are observed. For comparison, the particles are relatively uniform and small in Co-Re/Al₂O₃ owing to the promoted dispersion.^[12]

N₂ adsorption-desorption isotherms studies identified a little decrease in the pore volume and the pore size for Co-Re/Al₂O₃ (Table S1 and Figure S1). According to TPR profiles, Co₃O₄ is found to be reduced to metallic state through two distinct stages, i.e. the peaks located at about 370 °C and 450 °C, respectively attributed to reduction of Co₃O₄ to CoO and the transformation of CoO to metallic cobalt (Figure 4a). Similar to the reported works, Re favours the reducibility of the catalyst and the peaks for Co-Re/Al₂O₃ in TPR spectra negatively shift.^[13] Co 2p XPS spectra show peaks located at similar bind energies, which indicate the electron state of the cobalt species is not affected by Re (Figure 4b). Thus, we suggest that Re benefits the reduction degree of Co₃O₄ mainly depending on the H₂ spillover.

Influence of Promoter on Catalytic Performance

Structural diversity leads to a higher conversion rate for Co-Re/Al₂O₃ than for Co/Al₂O₃, thus the pressure of the batch reactor significantly decreased for Co-Re/Al₂O₃ in Figure 5a. It is expected

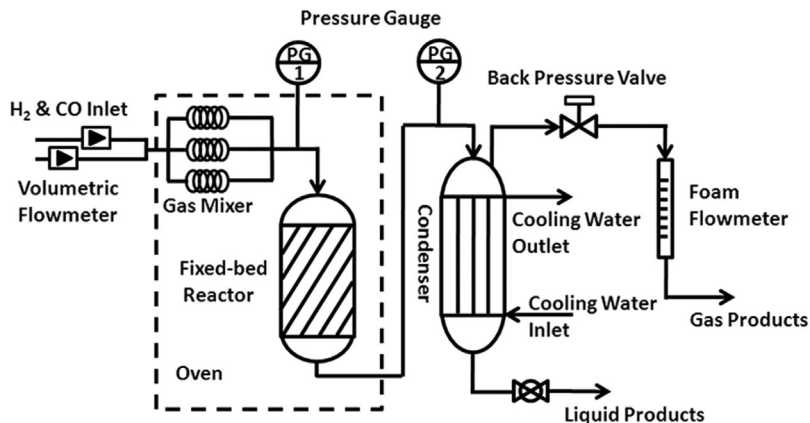


Figure 1. The fixed-bed micro-reactor system (Autoclave Engineering, BTR-Jr-PC).

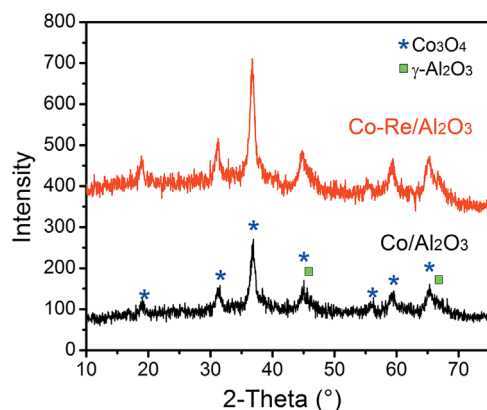


Figure 2. XRD spectra of calcined catalysts.

that the bifunctional FT catalyst, which combines cobalt and rhenium, might enhance C_{10} - C_{20} selectivity compared to that of Co/Al_2O_3 . In fact, activity experiments carried out in a batch reactor at 220 °C turn out to be similar to our conjecture (see Table 1 and Figure 5b). The result in Table 1 indicates that the promotion with Re efficiently increases the selectivity to C_{10} - C_{20} diesel fuels, and significantly reduces the heavy hydrocarbons. It is worth noting that straight chain hydrocarbons are the main products, and therefore the selectivity of hydrocarbons was calculated including the branched products without identification due to low contents. The C_{10} - C_{20} selectivity reached 57.5 % over $Co-Re/Al_2O_3$, which is higher than that of ASF distribution ($\alpha = 0.87$, maximum C_{10} - C_{20} selectivity 39.7 %).^[14] However, the catalytic activity still needs to increase and strategies must be applied to reduce the undesired products such as methane, light hydrocarbons, and wax to further increase the selectivity to C_{10} - C_{20} .

Yan et al.^[15] reported a continuous biomass to liquid process to produce aviation turbine fuels. The authors also studied the effects of temperature, reactor pressure, and GHSV on the FT catalytic performances. However, direct increase of high selectivity to diesel fuel or other desired hydrocarbons through tuning of operation condition was rarely reported. We therefore performed further studies of the $Co-Re/Al_2O_3$ system at low pressure by using a continuous fixed-bed reactor (Figure 1).

Influence of Operation Conditions on Catalytic Performance

Figure 6 shows the catalytic results over $Co-Re/Al_2O_3$. CO conversion exhibits expected trend versus system pressure up

to 1000 kPa. Nevertheless increasing in CO conversion turns to be slow after 850 kPa (Figure 6a). For FT hydrocarbon selectivity, catalyst worked at pressures higher than 700 kPa shows significantly enhanced C_{10} - C_{20} selectivity as shown in Figure 6b. Furthermore, the investigated conditions favour light liquid products with lower C_{21+} selectivity and higher CH_4 and C_2 - C_4 selectivities as compared to the ideal ASF distribution with the highest selectivity to the C_{10} - C_{20} hydrocarbons, thus revealing the hydrogenolysis of hydrocarbons during the FT synthesis. The hydrogenolysis favours the formation of CH_4 and shows a negative contribution to FT synthesis. Therefore, the efficient control of hydrogenolysis is expected to improve the diesel selectivity. On the other hand, the increasing pressure can also efficiently adjust the C_{10} - C_{20} selectivity to 58.7 % at 1000 kPa, which is 18.9 % higher than ASF selectivity. Figure 6c depicts the C_{5+} selectivity details. A narrow distribution suggests the high C_{10} - C_{20} selectivity, especially at 1000 kPa.

According to reference, the product capacity of BTL-FT plants is greatly reduced due to low CO conversion (< 25 %) at low pressures and the syngas ratio may further influence the FT synthesis process.^[16,17] It is expected to increase the catalytic activity through changing the H_2/CO ratio. Note that GHSV and temperature were kept constant (1000 h^{-1} , 200 °C), and pressure of 850 kPa was chosen based on the above results since the threshold CO conversion and high C_{5+} selectivity were obtained at this pressure. However, CO conversion increases slightly and linearly from 16.05 % to 20.56 % with increasing H_2 content in the syngas feed (Figure 7a). Furthermore, H_2/CO ratio significantly influences the selectivity to C_{21+} , while showing little effect on that to CH_4 , C_2 - C_4 , C_5 - C_9 , and C_{10} - C_{20} , as shown in Figure 7b. The C_{21+} selectivity decreases with an increase in H_2/CO ratio, while selectivities to other products (C_1 - C_{20}) increase. The total amount of C_1 - C_9 is equal to the ASF distribution. C_{10} - C_{20} selectivity is far higher than ideal ASF distribution whereas C_{21+} selectivity declined greatly, as detailed in Figure 7c. However, C_{10} - C_{20} and C_{21+} selectivities become H_2/CO ratio independent after inlet syngas ratio reaches 2.00. We speculate that increasing H_2 content probably accelerates the H_2 dissociation for the formation of CH_3 species, thus further influencing the chain termination process. In fact converted H_2/CO ratio initially increases and becomes stable after H_2/CO ratio reaches 2.00 (Figure S2).

To further study the influence of syngas on catalytic activity and product selectivities, GHSV was investigated. Pressure and temperature were still 850 kPa and 200 °C. The syngas ratio

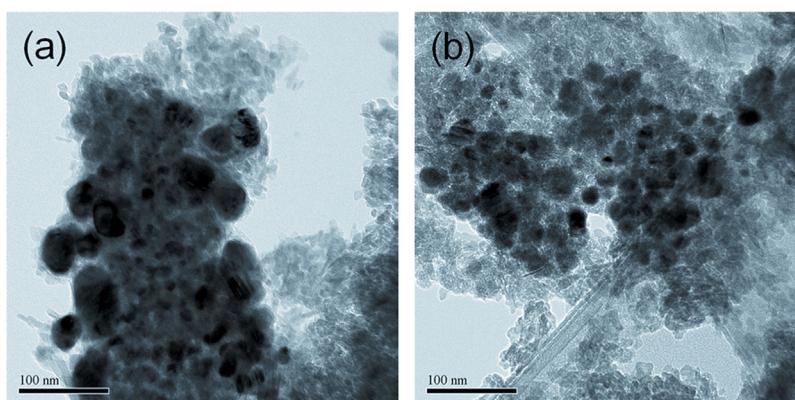


Figure 3. TEM images of (a) Co/Al_2O_3 and (b) $Co-Re/Al_2O_3$.

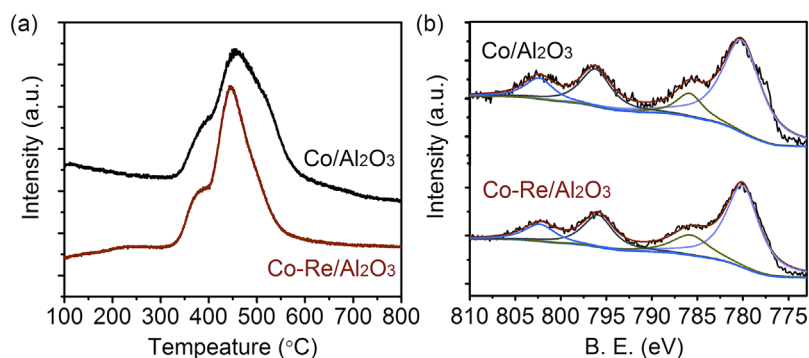


Figure 4. (a) TPR and (b) Co 2p XPS spectra of Co/Al₂O₃ and Co-Re/Al₂O₃.

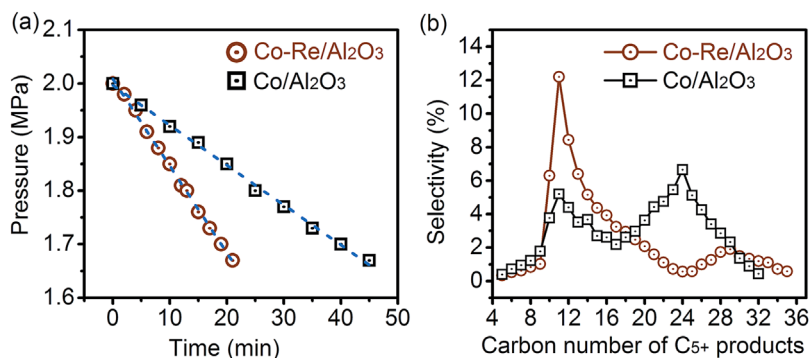


Figure 5. FT synthesis activity measured by batch-type autoclave. (a) Pressure variations with reaction time. (b) Comparison of C₅₊ product selectivities over Co/Al₂O₃ and Co-Re/Al₂O₃.

was optimized to 2.30 based on the highest CO conversion obtained at this value. Figure 8a shows that decreasing the GHSV, actually increasing the reactant residence time, significantly increases the CO conversion. In agreement with the literature, the product selectivity also depends on the syngas GHSV as shown in Figure 8b.^[18] Decreasing GHSV from 1300 h⁻¹ to 400 h⁻¹, CH₄ and C₂-C₄ selectivities decrease to 6.5 % and 14.1 %, respectively, whereas the C₁₀-C₂₀ and C₂₁₊ selectivity increases up to about 55.6 % and 11.9 %. However, selectivity to C₅-C₉ does not suffer. Though the selectivity to C₁₀-C₂₀ is far higher than the optimal selectivity calculated by the ideal ASF distribution for diesel production (Figure 8b and 8c), the undesirable products are still high. This unsatisfaction leads us to study the effects of GHSV. According to the above results, GHSV of 400 h⁻¹ was optimized due to high CO conversion and low C₁-C₉ selectivity.

Typically, cobalt based catalyst for FT synthesis is performed at temperatures between 200 °C and 240 °C.^[19,20] In fact, catalysis study at different temperatures, i.e. 185–230 °C, definitively solves two important problems: low CO conversion and high selectivity only to diesel. As shown in Figure 9a, CO conversion increases significantly with an increase in temperature up to 215 °C and then a moderate increase to 93.7 % at 230 °C is observed. Figure 9b depicts the influence of temperature on the hydrocarbon selectivities. Selectivity to CH₄, C₂-C₄, and C₂₁₊ decrease obviously, whereas C₅-C₉ and C₁₀-C₂₀ selectivities significantly improved. Compared to ideal ASF distribution, selectivity to CH₄ is still higher, but it is lower than 5 % at high temperature (>215 °C), which is industrially feasible. While C₂₁₊ selectivity decreases to 2.4 % at 215 °C and

it further decreases to 0.8 % at 230 °C, which may reduce the risk of pipe blockages in the industrial production. As shown in Figure 9c, we found that the distributions of C₅₊ products become narrow when catalysis temperature increases. The above results suggest that hydrocracking occurs and plays an important role in reducing C₂₁₊ production since hydrocracking of *n*-hexadecane was reported to enhance the C₅-C₁₁ selectivity.^[10] Selectivity to C₅-C₉ shows an obvious increase at high temperature, which confirms that hydrocracking and bond breaking probably happen in the middle C-C bond of C₂₁₊ hydrocarbons to form short chain compounds.

Compared with reported catalysts used in the mentioned reaction, as listed in Table 2, it is easy to see that publications mainly focused on catalytic performance or mechanism over

Table 1. FT synthesis over supported cobalt catalyst (a)

Catalysts	CO conversion (%)	Hydrocarbon selectivity (%) (b)				
		CH ₄	C ₂ -C ₄	C ₅ -C ₉	C ₁₀ -C ₂₀	C ₂₁₊
Co/Al ₂ O ₃	45.2	6.2	9.5	5.1	37.3	42.0
Co-Re/Al ₂ O ₃	57.6	9.8	12.5	3.3	57.5	16.9

(a) Reaction conditions: catalyst, 0.1 g; 220 °C; 2000 kPa; H₂/CO 2.00; 25 mL batch reactor; and time on steam 12 h. (b) The selectivity in the whole article was calculated on a molar carbon basis.

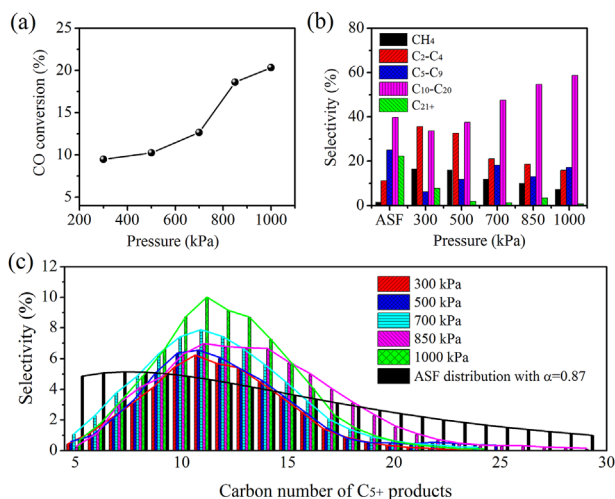


Figure 6. Influence of pressure on (a) CO conversion, (b) hydrocarbon selectivity, (c) selectivities to C₅+ products. The hydrocarbon selectivity from ASF distribution ($\alpha = 0.87$, optimum for C₁₀-C₂₀ production) is also displayed. Reaction conditions: catalyst (Co loading, 20 g/g; Re loading, 0.5 g/g), 0.18–0.25 mm in diameter, 2.0 g diluted with glass bead (2.5 mL, 0.15–0.18 mm in diameter); temperature 200 °C; H₂/CO ratio 2.00; GHSV based on catalyst volume 1000 h⁻¹; fixed-bed reactor; and time on steam 24 h for each condition.

cobalt-based catalysts at pressure of 200 kPa.^[11,21–27] Supported cobalt-based catalysts show great potential in producing C₅+ hydrocarbons with selectivity of 66–89 %. Beyond that, Wang et al. studied the catalytic performances over cobalt-based catalysts with various supports, e.g. Co/Na-meso-Y is highly selective to C₁₀-C₂₀ products, but the CO conversion is not attractive.^[11] Co/SBA-15 shows high FT activity and a selectivity of 40 % to C₁₀-C₂₀ hydrocarbons.^[22] Even though the CO conversion and selectivity are attractive, the reaction conditions

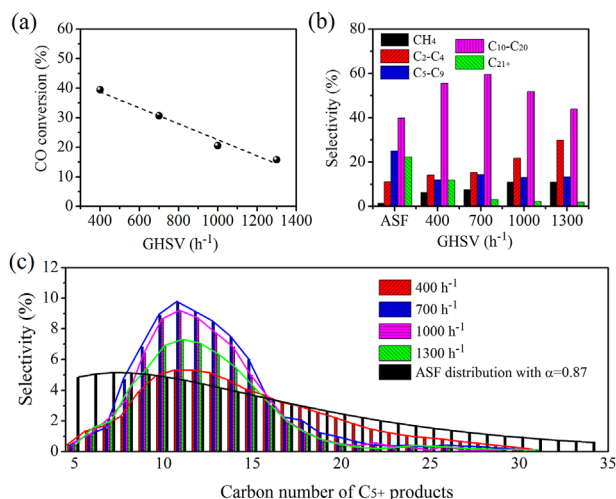


Figure 8. Influence of GHSV on (a) CO conversion, (b) hydrocarbon selectivity, and (c) selectivities to C₅+ products. The selectivity from ASF distribution is plotted as well. Reaction conditions: catalyst (Co loading, 20 g/g; Re loading, 0.5 g/g), 0.18–0.25 mm in diameter, 2.0 g diluted with glass bead (2.5 mL, 0.15–0.18 mm in diameter); pressure 850 kPa; temperature 200 °C; H₂/CO ratio 2.30; fixed-bed reactor; and time on steam 24 h for each condition.

are harsh. Holmen et al. reported Co-Re/Al₂O₃ catalyst with alkali impurities processed 82–83 % of C₅+ selectivity at 50 % CO conversion at relative mild conditions.^[23,24] However, the pressure applied is still high. Additionally, Campbell et al. studied the FT performance of Co-Rh/Nb₂O₅ under 100 kPa, but the CO conversion is not attractive.^[28] Similarly, Co-Ni-Cs/La₂O₃ catalyst exhibits low FT activity and C₅+ selectivity under low pressure.^[29] Accordingly, Co-Re/Al₂O₃ shows excellent selectivity to diesel fuels via FT synthesis and also attractive catalytic activity under investigated conditions.

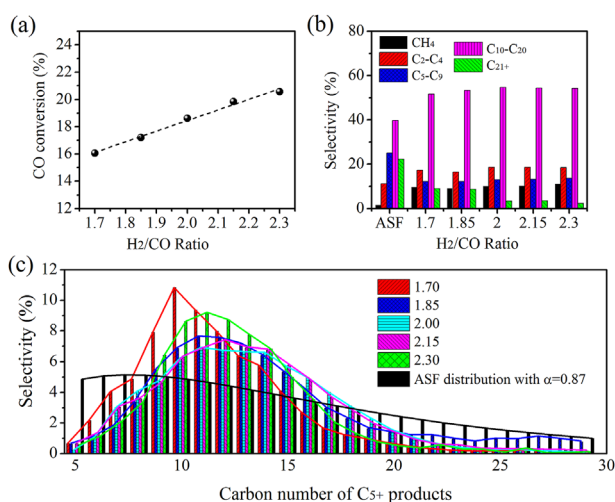


Figure 7. Influence of H₂/CO ratio on (a) CO conversion, (b) hydrocarbon selectivity, and (c) selectivities to C₅+ products. The selectivity from ASF distribution is depicted as well. Reaction conditions: catalyst (Co loading, 20 g/g; Re loading, 0.5 g/g), 0.18–0.25 mm in diameter, 2.0 g diluted with glass bead (2.5 mL, 0.15–0.18 mm in diameter); pressure 850 kPa; temperature 200 °C; GHSV based on catalyst volume 1000 h⁻¹; fixed-bed reactor; and time on steam 24 h for each condition.

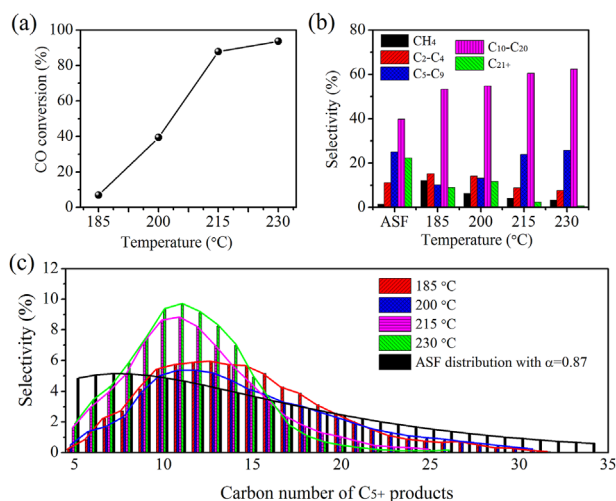


Figure 9. Influence of temperature on (a) CO conversion, (b) hydrocarbon selectivity, and (c) selectivities to C₅+ products. The selectivity from ASF distribution is shown as well. Reaction conditions are kept constant except for temperature. Reaction conditions: catalyst (Co loading, 20 g/g; Re loading, 0.5 g/g), 0.18–0.25 mm in diameter, 2.0 g diluted with glass bead (2.5 mL, 0.15–0.18 mm in diameter); pressure 850 kPa; H₂/CO ratio 2.30; GHSV based on catalyst volume 400 h⁻¹; fixed-bed reactor; and time on steam 24 h for each condition.

Table 2. Catalytic performances of cobalt-based catalysts used for FT synthesis

	Catalyst	Loading (g/g)	Reaction conditions				Catalysis performance			Ref.
			T (°C)	P (kPa)	Flow rate (mL/min)	H ₂ /CO ratio	Conversion (%)	C ₅₊ selectivity (%)	C ₁₀ -C ₂₀ selectivity (%)	
1	Co-Re/Al ₂ O ₃	20Co0.5Re	230	850	17	2.3	94	89	62	This work
2	Co/Al ₂ O ₃	15	230	2000	20	1.0	39	89	41	Peng et al. ^[11]
3	Co/Na-meso-Y	15	230	2000	20	1.0	40	89	60	
4	Co/MCM-41	20	250	2000	47	2.0	50	66	27	Ohtsuka et al. ^[21]
5	Co/SBA-15	5	250	2000	20	2.0	92	76	40	Tang et al. ^[22]
6	Co-Re/Al ₂ O ₃	20Co0.5Re (0.000026Na)	210	2000	148	2.1	50	82	n.a.	Øyvind et al. ^[23]
7	Co-Re/Al ₂ O ₃	20Co0.5Re	210	2000	250	2.1	50	81	n.a.	Balonek et al. ^[24]
8	Co-Re/Al ₂ O ₃	20Co0.5Re (0.001Na)	210	2000	250	2.1	50	83	n.a.	
9	Co-Ru/Al ₂ O ₃	25Co0.27Ru	220	1500	75	2.1	94	80	n.a.	Ma et al. ^[25]
10	Co/Al ₂ O ₃	15	220	2000	300	2.0	35	82	n.a.	Griboval-Constant et al. ^[26]
11	Co-Ru/Al ₂ O ₃	20Co1Ru	240	2000	60	2.0	50	69	50	Sari et al. ^[27]
12	Co-Rh/Nb ₂ O ₅	1.9Co2.3Rh	150	100	20	2.0	26	68	n.a.	Frydman et al. ^[28]
13	Co-Ni-Cs/La ₂ O ₃	50Co20Ni1Cs	260	200	90	2.0	20	21	n.a.	Mirzaei et al. ^[29]

According to the catalytic results, a simple mechanism is suggested as shown in Figure 10. It is reported that Re can promote the reducibility of cobalt and then increase the CO hydrogenation rate of cobalt catalyst.^[30,31] The results obtained in batch reaction tests showed that Re can promote catalyst activity, which was in agreement with the literature. Re is much active in adsorbing and dissociating hydrogen and is reported to increase the reducibility of cobalt by hydrogen spillover.^[32] The activated hydrogen atoms tend to interact with CO adsorbed by adjacent Co particles and facilitate the formation of CH, CH₂, and CH₃ species.^[33] This process is of great value on the chain starting process and then actually increases the CO conversion.^[34] Therefore, hydrogen-rich syngas is suggested to provide more activated hydrogen atoms and then enhance the CO conversion, while carbon-carbon bonds formed with the above species may produce hydrocarbons of varying chain length. Figure 5 depicts the product distribution shifting to light liquid hydrocarbons over Co-Re/Al₂O₃ catalyst. High H₂/CO ratio favours dissociation of H₂ which favours the formation of CH₃, thus enhancing the chain termination. Furthermore, it has been demonstrated that the hydrocracking

of heavier hydrocarbons played important roles in enhancing the C₅-C₁₁ selectivity by using *n*-hexadecane as a model molecule.^[10] Based upon the combination of the above results and literatures,^[10,35] we therefore propose that hydrocracking of as-produced straight chain wax possibly happens at high temperature, thus increasing the C₁₀-C₂₀ selectivity.

CONCLUSIONS

In order to satisfy plasma-enhanced biomass gasification process and to make it more economical, we studied the FT synthesis at low pressures. Two cobalt-based catalysts, Co/Al₂O₃ and Co-Re/Al₂O₃, were compared. Re-promoted Co/Al₂O₃ showed better performance in reaction rate and liquid products distribution. This may be attributed to the fact that Re has a stronger ability in dissociation of hydrogen. 850 kPa appears to be a threshold for a satisfactory CO conversion along with appropriate diesel production and limited wax. Variations in syngas H₂/CO ratio have a moderate influence on CO conversion and hydrocarbon selectivity, with improved CO conversions and slightly changed C₅₊ selectivity observed at high H₂/CO ratios. CO conversion, C₅₊ selectivity, and diesel yield are promoted when GHSV is lowered due to longer residence time. High temperature also increases the catalyst activity, thus producing attractive CO conversion. The work reported herein can help provide a better understanding of FT synthesis for diesel production from low pressure syngas.

ACKNOWLEDGEMENTS

The authors are grateful for financial support provided by the Natural Sciences and Engineering Research Council of Canada.

REFERENCES

- [1] H. Schulz, *Appl. Catal. A-Gen.* **1999**, *186*, 3.
- [2] A. R. de la Osa, A. De Lucas, A. Romero, J. L. Valverde, P. Sanchez, *Catal. Today* **2011**, *176*, 298.
- [3] D. Leckel, *Energ. Fuel.* **2009**, *23*, 2342.

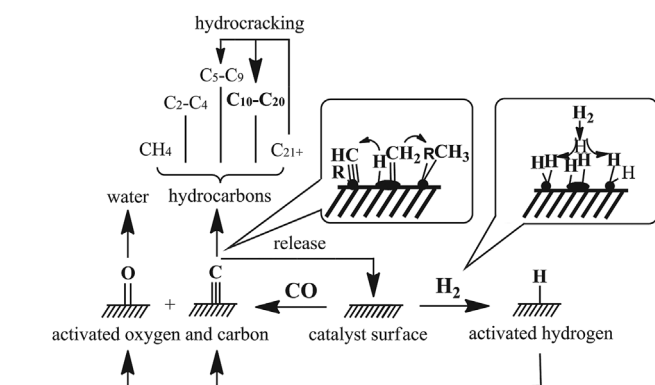


Figure 10. The role of noble metal promoter in the mechanisms for FT synthesis.

- [4] D. Unruh, K. Pabst, G. Schaub, *Energ. Fuel.* **2010**, *24*, 2634.
- [5] R. A. Dagle, J. A. Lizarazo-Adarme, V. L. Dagle, M. J. Gray, J. F. White, D. L. King, D. R. Palo, *Fuel Process. Technol.* **2014**, *123*, 65.
- [6] Q. Zhang, L. Dor, L. Zhang, W. Yang, W. Blasiak, *Appl. Energ.* **2012**, *98*, 219.
- [7] Y. Richardson, M. Drobek, A. Julbe, J. Blin, F. Pinta, *Recent Advances in Thermo-Chemical Conversion of Biomass*, Elsevier B.V., Boston **2015**.
- [8] M. Rønning, N. E. Tsakoumis, A. Voronov, R. E. Johnsen, P. Norby, W. V. Beek, B. Øyvind, E. Rytter, A. Holmen, *Catal. Today* **2010**, *155*, 289.
- [9] D. Kubička, R. Černý, *Ind. Eng. Chem. Res.* **2011**, *51*, 8849.
- [10] J. Kang, K. Cheng, L. Zhang, Q. Zhang, J. Ding, W. Hua, Y. Lou, Q. Zhai, Y. Wang, *Angew. Chem. Int. Edit.* **2011**, *50*, 5200.
- [11] X. Peng, K. Cheng, J. Kang, B. Gu, X. Yu, Q. Zhang, Y. Wang, *Angew. Chem. Int. Edit.* **2015**, *54*, 4553.
- [12] F. Diehl, A. Y. Khodakov, *Oil Gas Sci. Technol.* **2008**, *64*, 11.
- [13] C. L. Bianchi, *Catal. Lett.* **2001**, *76*, 155.
- [14] Q. Zhang, J. Kang, Y. Wang, *ChemCatChem* **2010**, *2*, 1030.
- [15] Q. Yan, Y. Fei, L. Jian, J. Street, J. Gao, Z. Cai, J. Zhang, *Bioresource Technol.* **2013**, *127*, 281.
- [16] S. L. Mok, M. J. Antal, *Thermochim. Acta* **1983**, *68*, 155.
- [17] K. Kim, Y. Kim, C. Yang, J. Moon, B. Kim, J. Lee, U. Lee, S. Lee, J. Kim, W. Eom, *Bioresource Technol.* **2013**, *127C*, 391.
- [18] M. Feyzi, A. Babakhanian, M. B. Gholivand, *Korean J. Chem. Eng.* **2014**, *31*, 37.
- [19] R. L. Espinoza, A. P. Steynberg, B. Jager, A. C. Vosloo, *Appl. Catal. A-Gen.* **1999**, *186*, 13.
- [20] A. Y. Khodakov, C. Wei, F. Pascal, *Chem. Rev.* **2007**, *107*, 1692.
- [21] Y. Ohtsuka, Y. Takahashi, M. Noguchi, T. Arai, S. Takasaki, N. Tsubouchi, W. Ye, *Catal. Today* **2004**, *89*, 419.
- [22] Q. Tang, P. Wang, Q. Zhang, Y. Wang, *Chem. Lett.* **2006**, *35*, 366.
- [23] B. Øyvind, S. Eri, E. A. Blekkan, S. Storsæter, H. Wigum, E. Rytter, A. Holmen, *J. Catal.* **2007**, *248*, 89.
- [24] C. M. Balonek, A. H. Lillebø, S. Rane, E. Rytter, L. D. Schmidt, A. Holmen, *Catal. Lett.* **2010**, *138*, 8.
- [25] W. Ma, G. Jacobs, Y. Ji, T. Bhatelia, D. B. Bukur, S. Khalid, B. H. Davis, *Top. Catal.* **2011**, *54*, 757.
- [26] D. Peña, A. Griboval-Constant, V. Lecocq, F. Diehl, A. Y. Khodakov, *Catal. Today* **2013**, *215*, 43.
- [27] A. Sari, *Chem. Eng. J.* **2014**, *244*, 317.
- [28] A. Frydman, D. G. Castner, C. T. Campbell, M. Schmal, *J. Catal.* **1999**, *188*, 1.
- [29] A. A. Mirzaei, B. Shirzadi, H. Atashi, M. Mansouri, *J. Ind. Eng. Chem.* **2012**, *18*, 1515.
- [30] E. Iglesia, S. C. Reyes, R. J. Madon, S. L. Soled, *Adv. Catal.* **1993**, *39*, 221.
- [31] A. M. Hilmen, D. Schanke, A. Holmen, *Catal. Lett.* **1996**, *38*, 143.
- [32] M. A. Ryashentseva, M. A. Ryashentseva, *Russ. Chem. Rev.* **1998**, *67*, 157.
- [33] E. L. Muetterties, J. Stein, *Chem. Rev.* **1979**, *79*, 479.
- [34] R. Zhang, X. Sun, B. Wang, *J. Phys. Chem. C* **2013**, *117*, 6594.
- [35] K. P. de Jong, J. Zečević, H. Friedrich, P. E. de Jongh, M. Bulut, S. van Donk, R. Kenmogne, A. Finiels, V. Hulea, F. Fajula, *Angew. Chem. Int. Edit.* **2010**, *49*, 10074.

Manuscript received November 8, 2016; revised manuscript received January 4, 2017; accepted for publication January 4, 2017.